Processor Lab — comparison report

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Created by Student Seba Joe Salloum · Processor Lab v4.0 (pipeline, cache and ISA teaching models) · 2026-11-09

Settings **Assumption**: 200 MHz, $40/hour; power OISC / SUBLEQ 0.25 W, MIPS32 (teaching subset) 0.45 W, RISC-V RV32I (teaching subset) 0.4 W, x86-64 (simplified subset) 0.8 W; cycle costs OISC / SUBLEQ sq=5; MIPS32 (teaching subset) ld=2/st=2/alu=1/br=1; RISC-V RV32I (teaching subset) ld=2/st=2/alu=1/br=1; x86-64 (simplified subset) ld=3/st=3/alu=1/alum=4/br=2.

Five-stage pipeline (MIPS32, RV32I): forwarding on, branch penalty 2 cycle(s) on every executed branch. Data cache: 64 B, 16 B blocks, direct-mapped, LRU, write-back, hit 1 / miss penalty 20 cycles.

Workload	OISC / SUBLEQ instr / multicycle cycles / pipeline cycles / ns / nJ (est.) / bytes / misses / AMAT	MIPS32 (teaching subset) instr / multicycle cycles / pipeline cycles / ns / nJ (est.) / bytes / misses / AMAT	RISC-V RV32I (teaching subset) instr / multicycle cycles / pipeline cycles / ns / nJ (est.) / bytes / misses / AMAT	x86-64 (simplified subset) instr / multicycle cycles / pipeline cycles / ns / nJ (est.) / bytes (est.) / misses / AMAT
Addition	3 / 15 / – / 75 / 18.8 / 36 / 2 / 5.44	4 / 7 / 9 / 35 / 15.8 / 16 / 1 / 7.67	4 / 7 / 9 / 35 / 14 / 16 / 1 / 7.67	3 / 10 / – / 50 / 40 / 8 / 1 / 7.67
Copy	3 / 15 / – / 75 / 18.8 / 36 / 2 / 5.44	2 / 4 / 7 / 20 / 9 / 8 / 1 / 11	2 / 4 / 7 / 20 / 8 / 8 / 1 / 11	2 / 6 / – / 30 / 24 / 5 / 1 / 11
Negation	2 / 10 / – / 50 / 12.5 / 24 / 1 / 4.33	3 / 5 / 8 / 25 / 11.3 / 12 / 1 / 11	3 / 5 / 8 / 25 / 10 / 12 / 1 / 11	3 / 7 / – / 35 / 28 / 7 / 1 / 11
Expression	4 / 20 / – / 100 / 25 / 48 / 2 / 4.33	6 / 10 / 10 / 50 / 22.5 / 24 / 1 / 6	6 / 10 / 10 / 50 / 20 / 24 / 1 / 6	4 / 14 / – / 70 / 56 / 11 / 1 / 6
Loop: sum 1..N	15 / 75 / – / 375 / 93.8 / 48 / 3 / 2.33	17 / 19 / 32 / 95 / 42.8 / 20 / 1 / 11	17 / 19 / 32 / 95 / 38 / 20 / 1 / 11	17 / 26 / – / 130 / 104 / 11 / 1 / 11
Condition: max, A≥B	6 / 30 / – / 150 / 37.5 / 108 / 2 / 3.22	5 / 8 / 12 / 40 / 18 / 24 / 1 / 7.67	6 / 9 / 15 / 45 / 18 / 28 / 1 / 7.67	4 / 12 / – / 60 / 48 / 13 / 1 / 7.67
Condition: max, B>A	8 / 40 / – / 200 / 50 / 108 / 2 / 2.67	6 / 9 / 13 / 45 / 20.3 / 24 / 1 / 7.67	5 / 8 / 12 / 40 / 16 / 28 / 1 / 7.67	5 / 15 / – / 75 / 60 / 13 / 1 / 6
Array sum: unrolled baseline	5 / 25 / – / 125 / 31.3 / 60 / 4 / 6.33	8 / 13 / 15 / 65 / 29.3 / 32 / 3 / 13	8 / 13 / 15 / 65 / 26 / 32 / 3 / 13	5 / 18 / – / 90 / 72 / 15 / 3 / 13
Array sum: looped (SUBLEQ indirect extension)	16 / 80 / – / 400 / 100 / 60 / 4 / 2.54	23 / 29 / 39 / 145 / 65.3 / 32 / 3 / 11	24 / 30 / 40 / 150 / 60 / 36 / 3 / 11	19 / 39 / – / 195 / 156 / 19 / 3 / 11
RV constant construction	2 / 10 / – / 50 / 12.5 / 24 / 2 / 7.67	3 / 4 / 7 / 20 / 9 / 12 / 1 / 21	3 / 4 / 7 / 20 / 8 / 12 / 1 / 21	3 / 5 / – / 25 / 20 / 11 / 1 / 21
Fibonacci	162 / 810 / – / 4050 / 1012.5 / 300 / 4 / 1.16	56 / 58 / 90 / 290 / 130.5 / 48 / 1 / 11	48 / 50 / 82 / 250 / 100 / 44 / 1 / 11	63 / 82 / – / 410 / 328 / 37 / 1 / 11
Array maximum	22 / 110 / – / 550 / 137.5 / 336 / 4 / 2.21	23 / 28 / 42 / 140 / 63 / 44 / 3 / 13	20 / 25 / 39 / 125 / 50 / 40 / 3 / 13	23 / 39 / – / 195 / 156 / 31 / 3 / 13

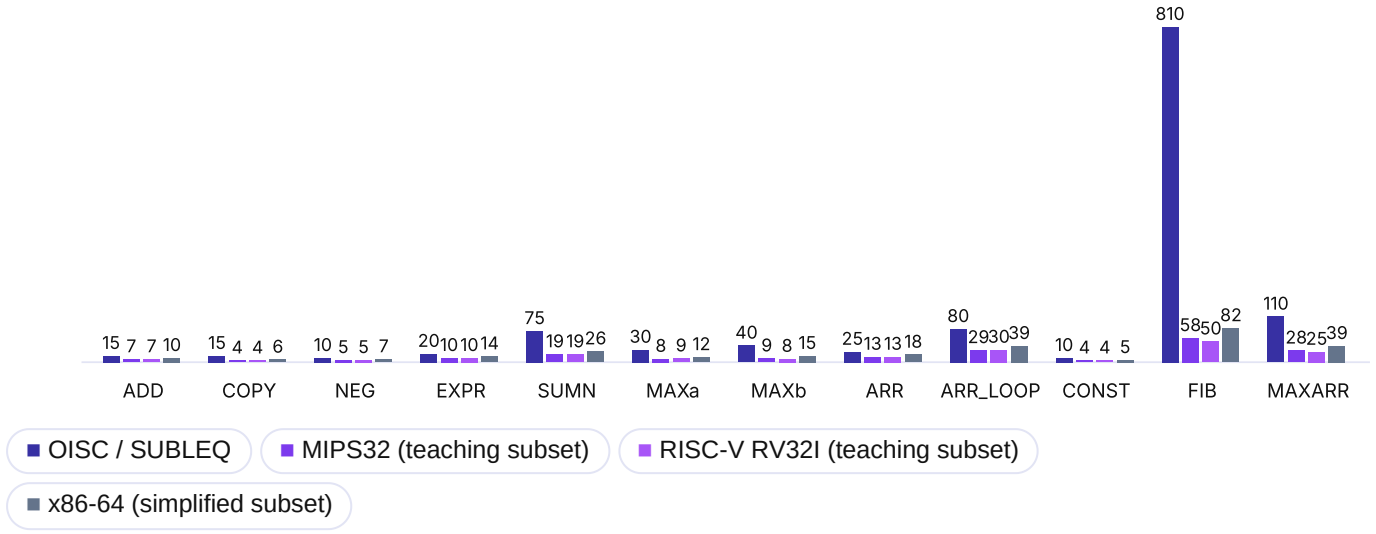
Methodology — four evidence classes:

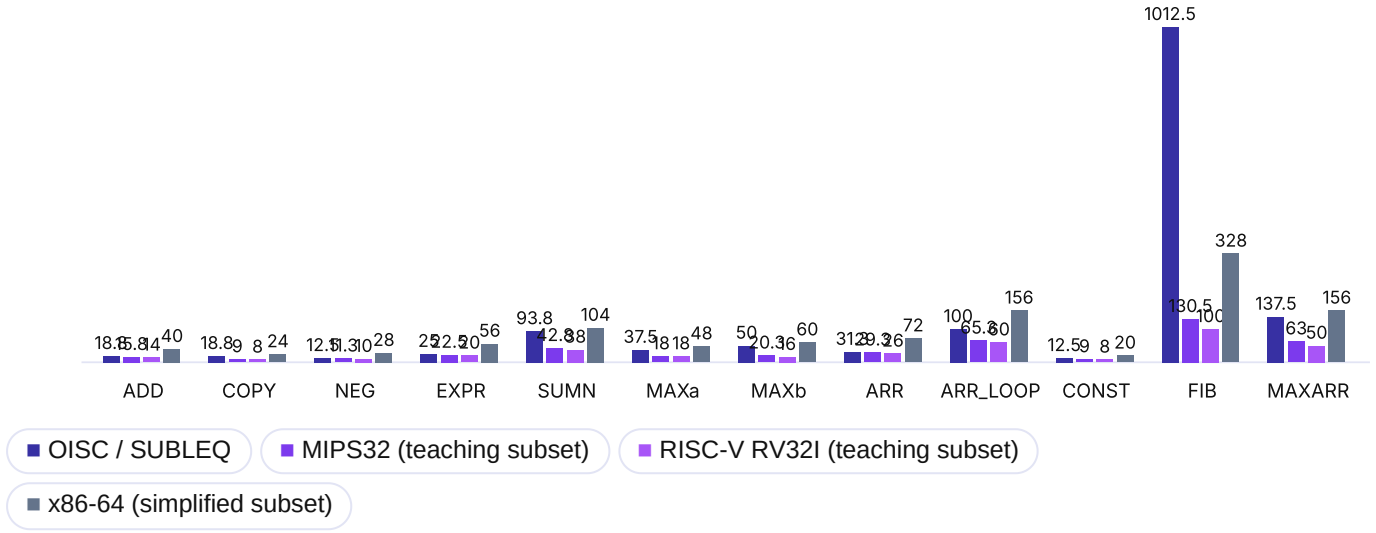
1. **ISA facts:** instruction formats, register rules, branch semantics and immediate layouts are taken from named architecture specifications. The RV32I base module is version 2.1; this report references the RISC-V Unprivileged ISA document version v20260120.
2. **Simulated (trace-derived) counts:** static and dynamic instruction counts, executed instruction mix, data reads/writes and accessed addresses come from the simulator's actual interpreter trace. Pipeline schedules are derived from the dynamic trace and the declared five-stage rules.
3. **Editable assumptions and calculated values:** cycle costs, clock rate, cache size/block size/associativity/replacement/write policy, hit time, miss penalty, forwarding and branch penalty are teaching-model inputs. Cycles, CPI, AMAT and time are *calculated* from simulated counts and those inputs; energy (assumed power × time) and labour cost (hours × rate) are *estimates*. None of them is a chip measurement.
4. **Unavailable hardware measurements:** silicon area, transistor count, physical clock behavior, measured power, real cache latency, branch predictor behavior and commercial implementation costs are not measured or inferred.

Cache and pipeline scope: set-associative (or direct-mapped) data cache with LRU/FIFO/seeded-random replacement, write-back or write-through; instruction fetches excluded; AMAT uses the single-level formula. Five-stage single-issue in-order pipeline for MIPS32 and RV32I with optional EX/MEM and MEM/WB forwarding, load-use stalls and a fixed branch penalty; no speculative execution, branch prediction or structural hazards. SUBLEQ looped array sum is explicitly an indirect-address extension; base SUBLEQ remains the unrolled baseline.

References: S1 RISC-V International, *The RISC-V Instruction Set Manual, Volume I: Unprivileged Architecture*, document v20260120 (January 2026), official specification; RV32I base ISA module v2.1. S2/S3 MIPS32 Architecture for Programmers, Vol. I/II. S4 Intel 64 and IA-32 Architectures Software Developer's Manuals. S5 SUBLEQ semantics and the explicitly stated simulator extension.

Charts: cycles and energy by workload





Defend every number

Number	Kind	Where to verify
Static and dynamic instructions, reads and writes, footprint	Simulated	Execution Simulator trace
Code size	Assumption	Encodings in Instruction Explorer (12-byte SUBLEQ teaching format; fixed 4 bytes MIPS32/RV32I; x86 approximate form sizes, not verified binary encodings)
Cycles, CPI, time, energy	Assumption	Counts from the trace multiplied by editable costs; time = cycles ÷ clock; energy = power × time
Development effort, labor cost	Assumption	Hours × hourly rate (educational model only)
RV32I has 40 base instructions	ISA fact	RISC-V specification (S1)

Area, transistors, prices	**Not measured**	No identified implementation